

Data and Database Management

Learning Objectives

- ▶ By the end of this lecture, students will be able to:
- ▶ Explain what a database is and why organizations use databases.
- ▶ Describe key components of Database Management Systems (DBMS).
- ▶ Understand database models (hierarchical, network, relational).
- ▶ Explain basic database operations (querying, updating, reporting).
- ▶ Recognize the importance of data quality, integrity, and security.

Database

- ▶ A **database** is an organized collection of related data stored electronically.
Allows:
- ▶ Easy storage
- ▶ Quick retrieval
- ▶ Efficient updating
- ▶ Secure access

Problems with Storing Data Without a Database

- ▶ Data redundancy (duplicates)
- ▶ Inconsistency
- ▶ Difficulty searching and updating
- ▶ Poor data security
- ▶ Limited sharing among departments

- ▶ A DBMS solves these problems.

DBMS

- ▶ A **Database Management System (DBMS)** is software used to define, create, manage, and manipulate databases.
- ▶ Examples:
- ▶ MySQL
- ▶ Oracle
- ▶ Microsoft SQL Server
- ▶ PostgreSQL
- ▶ Access

Functions of a DBMS

- ▶ **Data definition** (creating tables, schemas)
- ▶ **Data manipulation** (insert, update, delete)
- ▶ **Data retrieval** (SQL queries)
- ▶ **Data security and access control**
- ▶ **Backup and recovery**
- ▶ **Concurrency control** (multiple users accessing data)

Components of a DBMS

- ▶ **Data Definition Language (DDL)** - create/modify tables
- ▶ **Data Manipulation Language (DML)** - query and update data
- ▶ **Query Processor**
- ▶ **Database Administrators (DBA) tools**

Database Models

- ▶ **1. Hierarchical Model**
 - ▶ Tree-like structure
 - ▶ Parent-child relationships
- ▶ **2. Network Model**
 - ▶ More connections than hierarchical
 - ▶ Many-to-many relationships
- ▶ **3. Relational Model (most common today)**
 - ▶ Data stored in tables (rows and columns)
 - ▶ Uses **keys** to link tables
 - ▶ Based on mathematical relations
 - ▶ SQL used for queries

Relational Database Concepts

- ▶ **Table:** collection of related records
- ▶ **Row (record):** one instance of data
- ▶ **Column (field):** attribute of data
- ▶ **Primary key:** uniquely identifies a record
- ▶ **Foreign key:** links two tables
- ▶ **Relationships:** one-to-one, one-to-many, many-to-many

Database Design Concepts

- ▶ **1. Entity**
 - ▶ Something that we store data about (student, customer, product)
- ▶ **2. Attribute**
 - ▶ Property of an entity (name, phone, price)
- ▶ **3. Relationship**
 - ▶ Association between entities (student enrolls in course)
- ▶ **4. ER Diagrams (Entity-Relationship)**
 - ▶ Used for designing databases visually.

Data Quality

- ▶ High-quality data must be:
 - ▶ Accurate
 - ▶ Complete
 - ▶ Consistent
 - ▶ Timely
 - ▶ Valid
- ▶ Poor data quality → poor decisions.

Data Security and Privacy

- ▶ Databases must protect data from:
- ▶ Unauthorized access
- ▶ Malware
- ▶ Data breaches
- ▶ Errors and corruption
- ▶ Methods:
- ▶ User authentication
- ▶ Access control
- ▶ Encryption
- ▶ Backup and recovery plans

Database Roles in Organizations

- ▶ **Database Administrator (DBA)**
 - ▶ Manages and secures databases
 - ▶ Ensures performance and availability
- ▶ **Data Analyst**
 - ▶ Uses data for insights
- ▶ **Data Engineer**
 - ▶ Designs and optimizes data pipelines
- ▶ **Software Developers**
 - ▶ Build applications that use databases

Summary

- ▶ Data is a vital organizational resource
- ▶ Databases store and manage data efficiently
- ▶ DBMS enables security, accuracy, and easy access
- ▶ Relational databases and SQL are widely used
- ▶ Data quality, security, and design are essential

Questions

- ▶ Why are databases important to organizations?
- ▶ What problems arise without a DBMS?
- ▶ Which database model is the most widely used today and why?
- ▶ Why is data security important?



Systems Development

- ▶ Systems development is the process of **creating or modifying information systems** to meet business needs.
- ▶ Includes:
 - ▶ Planning
 - ▶ Analyzing requirements
 - ▶ Designing solutions
 - ▶ Developing systems
 - ▶ Implementing and maintaining systems

Why Do Organizations Develop Systems?

- ▶ Improve efficiency
- ▶ Automate tasks
- ▶ Enhance decision-making
- ▶ Solve existing problems
- ▶ Integrate new technologies
- ▶ Gain competitive advantage
- ▶ Meet customer or market needs

Types of Information Systems to Develop

- ▶ Transaction Processing Systems (TPS)
- ▶ Management Information Systems (MIS)
- ▶ Decision Support Systems (DSS)
- ▶ ERP Systems
- ▶ Mobile applications
- ▶ E-commerce platforms

The Systems Development Life Cycle (SDLC)

- ▶ Traditional method for developing systems, consisting of 6-7 phases:
- ▶ **Planning**
- ▶ **Analysis**
- ▶ **Design**
- ▶ **Development**
- ▶ **Testing**
- ▶ **Implementation**
- ▶ **Maintenance**

SDLC Phase 1 - Planning

- ▶ Key activities:
 - ▶ Identify problems/opportunities
 - ▶ Define project scope
 - ▶ Assess feasibility:
 - ▶ Technical
 - ▶ Economic
 - ▶ Operational
 - ▶ Schedule
 - ▶ Legal
 - ▶ Develop a project plan
- ▶ Deliverable: **Feasibility Study**

SDLC Phase 2 - Analysis

- ▶ Activities:
- ▶ Study current system (“As-Is” process)
- ▶ Identify user requirements
- ▶ Develop alternative solutions
- ▶ Logical models (DFDs, Use Cases)
- ▶ Deliverable: **System Requirements Specification**

SDLC Phase 3 - Design

- ▶ Activities:
 - ▶ Design system architecture
 - ▶ Database design (ERD → schema)
 - ▶ User interface design
 - ▶ System interfaces
 - ▶ Security plan
- ▶ Deliverable: **System Design Document**

SDLC Phase 4 - Development

- ▶ Activities:
 - ▶ Write code
 - ▶ Configure hardware/software
 - ▶ Build database
 - ▶ Integrate system components
- ▶ Deliverable: **Working system**

SDLC Phase 5 - Testing

- ▶ Types of testing:
 - ▶ Unit testing
 - ▶ Integration testing
 - ▶ System testing
 - ▶ User acceptance testing (UAT)
 - ▶ Ensures the system meets all requirements.

SDLC Phase 6 - Implementation

- ▶ Methods:
 - ▶ Direct cutover
 - ▶ Parallel implementation
 - ▶ Pilot implementation
 - ▶ Phased conversion
 - ▶ Ensure user training and data migration.

SDLC Phase 7 - Maintenance

- ▶ Types:
- ▶ Corrective (fix bugs)
- ▶ Adaptive (update system with new needs)
- ▶ Perfective (enhance features)
- ▶ Preventive (improve system performance)
- ▶ Maintenance consumes **60-80%** of system lifecycle cost.



Feasibility Analysis

Learning Objectives

- ▶ By the end of this lecture, students will be able to:
- ▶ Define feasibility analysis and explain its purpose
- ▶ Describe the main types of feasibility
- ▶ Conduct a basic feasibility study
- ▶ Explain why feasibility analysis is essential for system success

Feasibility Analysis

- ▶ **Feasibility analysis** is a preliminary study that evaluates whether a proposed system or project is **practical, achievable, and worthwhile** before major resources are committed.
- ▶ It determines:
- ▶ *Should we proceed?*
- ▶ *Can the project succeed technically, economically, and operationally?*

Purpose of Feasibility Analysis

- ▶ Avoid investing in unsuccessful projects
- ▶ Ensure requirements are realistic
- ▶ Estimate project costs, timelines, and benefits
- ▶ Identify risks and constraints early
- ▶ Support decision-making and approval
- ▶ Increase project success rate

Types of Feasibility (The 6 Key Categories)

- 1 Technical Feasibility
- 2 Economic Feasibility
- 3 Operational Feasibility
- 4 Schedule Feasibility
- 5 Legal Feasibility
- 6 Organizational / Political Feasibility

► These categories ensure a complete assessment of the project.

Technical Feasibility

- ▶ Evaluates whether the organization has or can acquire the **technology, skills, hardware, and software** required.
- ▶ Questions to consider:
 - ▶ Do we have the technical expertise?
 - ▶ Does required technology exist?
 - ▶ Are infrastructure upgrades needed?
 - ▶ Are security requirements achievable?
- ▶ Deliverable: **Technology Assessment Report**

Economic Feasibility (Cost-Benefit Analysis)

- ▶ Determines whether the project is **financially worthwhile**.
- ▶ Costs include:
 - ▶ Hardware & software purchases
 - ▶ Development and testing
 - ▶ Training
 - ▶ Maintenance
 - ▶ Staffing
- ▶ Benefits include:
 - ▶ Reduced costs
 - ▶ Increased revenue
 - ▶ Improved productivity
 - ▶ Better decision-making
 - ▶ Customer satisfaction
- ▶ Typical methods:
 - ▶ ROI (Return on Investment)
 - ▶ NPV (Net Present Value)
 - ▶ Payback period
- ▶ Deliverable: **Cost-Benefit Analysis**

Operational Feasibility

- ▶ Assesses whether the project will operate successfully in the organization after implementation.
- ▶ Key considerations:
- ▶ Will users accept the system?
- ▶ Will it solve the problem?
- ▶ Does it fit organizational culture?
- ▶ Are workflows aligned?
- ▶ Is training adequate?
- ▶ Most system failures happen due to **operational issues**, not technical ones.

Schedule Feasibility

- ▶ Evaluates whether the project can be completed within the given time frame.
- ▶ Questions:
 - ▶ Is the deadline realistic?
 - ▶ Are resources available on time?
 - ▶ Are there time-sensitive requirements (legal, market, seasonal)?
- ▶ Tools:
 - ▶ Gantt charts
 - ▶ PERT

Legal Feasibility

- ▶ Ensures that the project complies with legal and regulatory requirements.
- ▶ Check for:
 - ▶ Data protection laws
 - ▶ Copyright and licensing
 - ▶ Contractual constraints
 - ▶ Industry regulations
 - ▶ Privacy and security policies

Organizational / Political Feasibility

Assesses how the project aligns with:

- ▶ Organizational goals
- ▶ Internal politics
- ▶ Management support
- ▶ Stakeholder interests
- ▶ Questions:
 - ▶ Do managers support this project?
 - ▶ Does it conflict with other departments?
 - ▶ Will employees resist change?

The Feasibility Study Report

A complete feasibility report includes:

- ▶ Executive summary
- ▶ Problem definition
- ▶ Proposed solution
- ▶ Technical feasibility
- ▶ Economic feasibility
- ▶ Operational feasibility
- ▶ Schedule feasibility
- ▶ Legal feasibility
- ▶ Organizational feasibility
- ▶ Recommendation (Go/No Go)

Example - Feasibility for E-Commerce System

- ▶ **Project:** Online store for a retail chain
- ▶ **Summary:**
- ▶ **Technical:** Cloud technology available → feasible
- ▶ **Economic:** Expected ROI 25% → feasible
- ▶ **Operational:** Staff must be trained → partially feasible
- ▶ **Schedule:** 6-month deadline → tight but possible
- ▶ **Legal:** Must comply with payment security standards → feasible
- ▶ **Organizational:** High management support → feasible
- ▶ **Recommendation:** Proceed with modifications and training plan.

Why Projects Fail Without Feasibility Analysis

- ▶ Underestimated costs
- ▶ Unrealistic schedules
- ▶ Lack of user acceptance
- ▶ Legal conflicts
- ▶ Technical barriers
- ▶ Poor alignment with business goals
- ▶ Feasibility analysis prevents these issues early.

Advantages of Feasibility Analysis

- ▶ Saves time and money
- ▶ Improves project planning
- ▶ Identifies risks early
- ▶ Encourages user involvement
- ▶ Helps in informed decision-making

Questions

- ▶ Which type of feasibility is most important? Why?
- ▶ Can a project with low economic feasibility still be approved?
- ▶ Why do some organizations skip feasibility studies?

Project Planning

Learning Objectives

- ▶ By the end of this lecture, students will be able to:
- ▶ Explain the purpose of project planning
- ▶ Identify the components of a project plan
- ▶ Describe key tools used in project planning
- ▶ Understand risk planning and resource planning

Project Planning

- ▶ Project planning is the process of defining:
 - ▶ **What** needs to be done
 - ▶ **How** it will be done
 - ▶ **Who** will do it
 - ▶ **When** it will be done
 - ▶ **How much** it will cost
- ▶ Planning is conducted after project initiation and before execution.

Importance of Project Planning

- ▶ Effective planning:
- ▶ Reduces uncertainty
- ▶ Improves team coordination
- ▶ Provides a roadmap for execution
- ▶ Helps manage time, cost, and quality
- ▶ Ensures resource availability
- ▶ Identifies potential risks early

Key Components of a Project Plan

- ▶ A complete project plan typically includes:
- ▶ **Scope statement**
- ▶ **Work Breakdown Structure (WBS)**
- ▶ **Schedule (Gantt, PERT, CPM)**
- ▶ **Budget estimate**
- ▶ **Resource plan**
- ▶ **Risk management plan**
- ▶ **Quality plan**
- ▶ **Communication plan**
- ▶ **Procurement plan (if needed)**

Project Scope Planning

- ▶ The scope defines:
- ▶ Project objectives
- ▶ Deliverables
- ▶ Boundaries
- ▶ Constraints
- ▶ Assumptions
- ▶ **Scope creep** occurs when uncontrolled changes expand a project's scope.

Project Planning Process

- ▶ Define scope
- ▶ Create Work Breakdown Structure (WBS)
- ▶ Identify activities
- ▶ Sequence activities
- ▶ Estimate time and costs
- ▶ Develop schedule
- ▶ Assign resources
- ▶ Identify and assess risks
- ▶ Create full project plan

Why Projects Fail Without Proper Planning

- ▶ Unclear goals
- ▶ Poor scheduling
- ▶ Underestimated workload
- ▶ Lack of resources
- ▶ No risk management
- ▶ Poor communication
- ▶ Budget overruns
- ▶ Planning prevents these issues.

Questions

- ▶ Why is WBS important for project planning?
- ▶ Why do some teams skip planning?
- ▶ What is the difference between Gantt and PERT charts?